

1. User Documentation	2
1.1 Copyright & Warranty Information	3
1.2 Table of Contents.	4
1.3 Table of figures.	5
1.4 1.0 Introduction.	6
1.4.1 1.1 System Overview.	7
1.4.2 1.2 Audience Description.	8
1.4.3 1.3 Applicability Statement.	9
1.4.4 1.4 Purpose Statement.	10
1.4.5 1.5 Document Usage Description.	11
1.4.6 1.6 Conventions.	12
1.4.7 1.7 Change log.	13
1.5 2.0 Introductory Kit	14
1.6 3.0 Getting Started Guide	15
1.7 4.0 User Manual	19
1.7.1 4.1 User Guide (Android)	20
1.7.2 4.2 User Guide (iOS)	27
1.8 5.0 System Administrator Guide	33
1.9 6.0 Installation Guide	36
1.9.1 6.1 Install App	37
1.9.2 6.2 Install Hardware Node	38
1.10 7.0 Special Notes	42
1.10.1 APPENDIX_a Statement of authorship	43
1.10.2 Bibliography	45
1.10.3 Glossary of Terms	46

User Documentation

PARK ME @ LA TROBE

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Table of Contents.

Copyright and Warranty Information

TABLE OF CONTENTS

Table of figures

1.0 Introduction

1.1 System Overview

1.2 Audience Description

1.3 Applicability Statement

1.4 Purpose Statement

1.5 Document Usage Description

1.6 Conventions

1.7 Change Log

2.0 Introductory Kit

3.0 Getting Started Guide

4.0 User Manual

4.1 User Guide (Android)

4.2 User Guide (iOS)

5.0 System Administrator Guide

6.0 Installation Guide

6.1 Install App

6.2 Install Hardware Node

7.0 Special Notes

APPENDIX_a Statement of authorship

Bibliography

Glossary of Terms

Table of figures.

Figure	Description
Fig. 1.1.1	System overview displayed as a block diagram
Figure 3.0.1	Buttons and icons
Figure 4.1.1	Home page (Android)
Figure 4.1.2	Refresh button on home page (Android)
Figure 4.1.3	Quick Links Home Page (Android)
Figure 4.1.4	Building List (Android)
Figure 4.1.5	Search Building List (Android)
Figure 4.1.6	Quick Links Home Page (Android)
Figure 4.1.7	Choose Car Park (Android)
Figure 4.1.8	Quick Links Home Page (Android)
Figure 4.1.9	Selected car Park with building (Android)
Figure 4.1.10	Selected Car Park without building (Android)
Figure 4.1.11	Quick Links Home Page (Android)
Figure 4.1.12	Information Daily Tickets (Android)
Figure 4.1.13	Information Visitors (Android)
Figure 4.1.14	Quick Links Home Page (Android)
Figure 4.1.15	Campus Map (Android)
Figure 4.1.16	Map Options (Android)
Figure 4.2.1	Home page(iOS)
Figure 4.2.2	Quick Links Home Page(iOS)
Figure 4.2.3	Building List(iOS)
Figure 4.2.4	Search Building List(iOS)
Figure 4.2.5	Quick Links Home Page(iOS)
Figure 4.2.6	Choose Car Park(iOS)
Figure 4.2.7	Quick Links Home Page(iOS)
Figure 4.2.8	Selected car Park with building(iOS)
Figure 4.2.9	Selected car Park without building (iOS)
Figure 4.1.10	Quick Links Home Page(iOS)
Figure 4.1.11	Information Daily Tickets(iOS)
Figure 4.1.12	Information Visitors (iOS)
Figure 4.1.13	Quick Links Home Page (iOS)
Figure 4.1.14	Campus Map (iOS)
Figure 4.1.15	Map Options(iOS)
Figure 5.0.1	Database log files web page
Figure 5.0.2	hover over bar CP2 example.

1.0 Introduction.

This section contains the following pages:

[1.1 System Overview.](#)

[1.2 Audience Description.](#)

[1.3 Applicability Statement.](#)

[1.4 Purpose Statement.](#)

[1.5 Document Usage Description.](#)

[1.6 Conventions.](#)

[1.7 Change log.](#)

1.1 System Overview.

The Park Me @ La Trobe app will be an app for students, staff and visitors to Latrobe University to find a parking spot on campus. The app can be accessed on devices such as android and iOS. The app would be supported by camera nodes around the university to detect free car spots. The user can choose a spot based on the best available, by car park or by closest building.

The system displayed as a block diagram looks as follows:

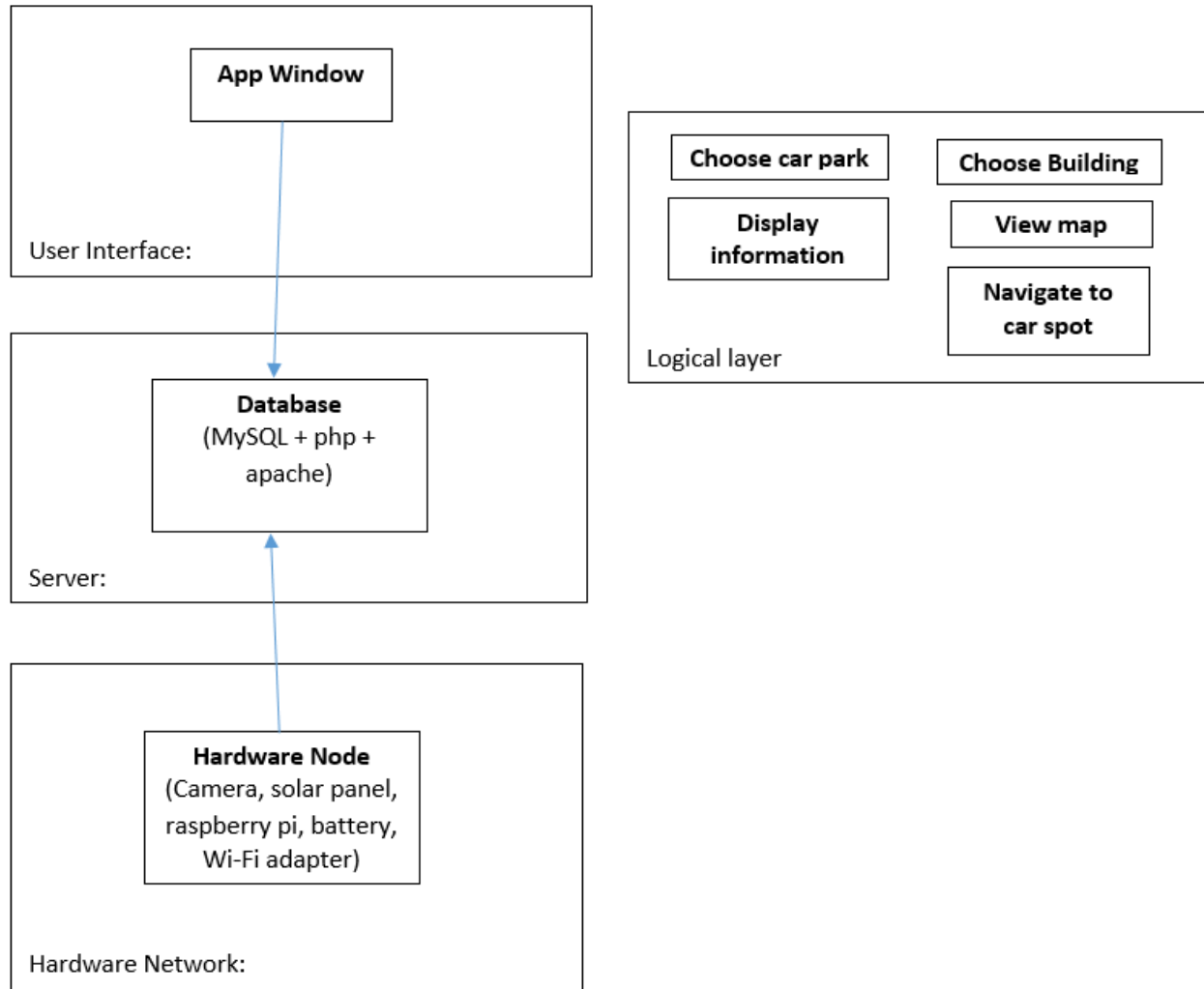


Figure 1.1.1 System overview displayed as a block diagram

1.2 Audience Description.

The users of our system will be:

- (general) User: This includes anyone but more specifically students, staff and visitors to La Trobe University. Basic phone knowledge is required to download, operate and interact with the app. Users are also expected to have a basic understanding of how gps navigation on a phone works. Users are also expected to have a phone with wi-fi or internet and GPS.
- Admin: This refers to the IT department staff who will maintain the system. They are expected to have a basic understanding of retrieving data from server.

1.3 Applicability Statement.

Hardware Node:

The hardware nodes are running Raspbian, a Linux distribution for embedded computers and was supported with the following technologies:

OpenCV 3.0, Python: 2.7.3 and Netbeans 8.0.2.

Backend server:

The backend server is running:

Apache 2.4.12, MySQL 5.6.24, PHP 5.6.8.

Android/iOS App:

- A single, cross-platform app is being developed with Appcelerator Studio 4.3.0.
- It is being built with Titanium SDK 5.0 and Alloy MVC framework, Android SDK and XCode.
- The minimum version of Android support is 4.0 (API 14) and the target version is 5.0 (API 21).
- The Android app is being developed for and tested on a Nexus 5.
- The iOS app is being developed and tested on a iPhone 5S.

1.4 Purpose Statement.

The main purpose of the system is to provide an app to navigate a user to find a car spot at La Trobe.

Major applications of the app are:

- Provide information to new users about parking at La Trobe.
- Provide a way for users to choose the way the app picks a car spot.
- Reduce the amount of time a user spends seeking an available car spot at La Trobe.
- Provide a way to find the closest car spot.

1.5 Document Usage Description.

This document has been divided into 7 sections, defined as follows:

1. Introduction:

This section provides an overall description of the audience, purpose, application of the project. Also provides an overall description of this document and the conventions used.

2. Introductory Kit

This section provides a summary of all the features available to users and administrators.

3. Getting Started Guide

This section provides a step by step guide to help users get started with the system. It also covers basic features of the system. Includes description of the layout and icons of the app.

4. User Guide

This section provides a step by step guide showing how to access and use all the user features of the system.

5.0 System Administrator Guide

This section provides a step by step guide showing how to access and use all the administrator features of the system.

6.0 Installation Guide

This section provides a step by step guide on how to install and set up the app on iOS and android. Also the hardware for installation in the car parks.

7.0 Special Notes

This section includes supporting documentation such as glossary of terms, appendices and bibliography.

1.6 Conventions.

The diagrams in this document were created with draw.io and follow the standards of UML.

The following language conventions were used in this document:

- Car Park refers to the Whole car park such as car park 1 and car park 2a
- Car Spot refers to the individual parking spot.
- When the user 'Clicks' this refers to a single tap on the screen
- Figure codes follow the following format: Fig. <page number>.<image number> e.g. Fig. 4.2.3.1

There may be unfamiliar terms used in this document and they have been described in the [Glossary](#).

1.7 Change log.

Date	Changes	Changed by
	First release: Document Layout	Kylie Marshall
10 August 2015	Added Content to: Copyright & Warranty Information, Table of Contents Table of figures Section 1: Introduction	Kylie Marshall
August 2015	Added content to: 2.0 Introductory kit 3.0 Getting Started Guide 4.1 user guide(Android) 4.2 user guide(iOS)	Kylie Marshall
18 Sep 2015	Added Content to: 3.0 Getting Started Guide 6.1 Install app	Aleks Zuchowicz
21 Sep 2015	Reordered and renaming of section 4.	Kylie Marshall
25 Sep 2015	Added content to Table of figures	Kylie Marshall
1 Oct 2015	added updated log files web page to 5.0 System administrator guide	Kylie Marshall
6 Oct 2015	Updated content on: 3.0 Getting Started Guide 4.1 user guide(Android) 4.2 user guide(iOS) 5.0 Administrator guide	Kylie Marshall
12 Oct 2015	Spelling and grammar corrections through document. Updated Glossary.	Paul Dwyer

2.0 Introductory Kit

System features summary

A regular user of the app should be able to:

- View the amount of available spots in each car park
- Navigate to a car spot in a chosen car park
- Navigate to a car spot closest to chosen building
- Navigate to the closest, best available car spot
- View information about parking at La Trobe
- View overall car park map
- Change sound and other settings

A admin should be able to:

- To use the app as described above
- View log files generated by the app and the hardware nodes
- To view useful statistics on car park behaviour.

3.0 Getting Started Guide

This guide contains information on where to start, how to get the app and an explanation of buttons and icons used in the app. It is recommended you read through sections one and two before reading this section.

Where to start?

You will need to have the app on your phone or tablet. The app can be downloaded from the Google Play Store for Android devices, or the App Store for iOS devices. For more information on how to install the app please see the installation guide: [6.1 Install App](#).

Which devices does the app support?

- Any device running Android 4.3 or above
- Any device running iOS 7.1 or above

Do I need any additional apps?

Yes. For the maps to display on Android, the latest version of Google Play services is required. It is also highly recommended to have Google Maps and Apple Maps up to date for turn by turn navigation. iOS users have an option of using Google Maps over Apple Maps for turn by turn navigation. In this case, Google Maps must be installed.

What skills do I need?

To use this app and understand the guide it is expected that users know how to tap, swipe, open an app and close an app.

Do I need the internet to use this app?

Yes. This app requires an internet connection to refresh the number of spots available. Some features of the app such as information do not require an internet connection. It is recommended to use the app when close to La Trobe and hence you can use the La Trobe Wi-Fi to connect to the internet.

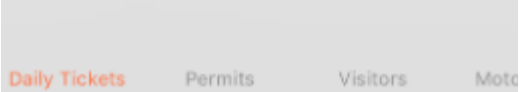
Do I need to log in or sign up for anything?

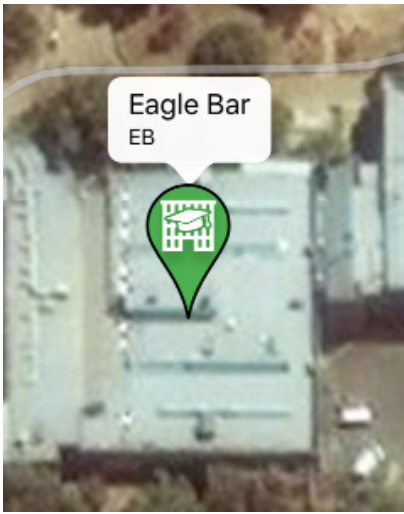
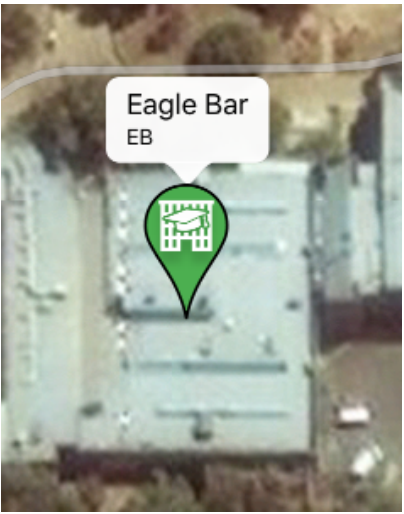
No. There is no log in or sign up required to use this app.

Buttons and icons

These descriptions may help you understand how to navigate through the app. The name and description are included. Note some devices may have a different icon but should be similar to the ones presented.

Name	Description	Android icon	iOS icon
Back button	This button will close the page you are currently on. It will return you to the page you were previously on. Continuous clicking will return you to the home page. Cannot return from the home page. Button located in the top left corner.		
Forward button	This icon indicates that the row is clickable. Clicking on the icon will open a new page.		

Refresh button	Used to refresh the number of car spots available for each car park. Button located on the home page in the top right corner.		
Slider	Sliders change a feature between on and off. Blue indicates on and grey indicates off. The feature that is changed is described in text next to the slider.		
Search button	This button opens the search bar in building list. Only on android. The search on iOS is at the top of the list.		NA
Add button	This button is located on the campus map page. It allows the user to add or remove features shown on the map.		
Tab bar	This is a feature that may change between devices. In general the tab that is being viewed is highlighted. A different tab can be selected by tapping on its name (or swiping right or left on android only).		
Info button	This button when clicked will open a new page and display information about parking at La Trobe		
Map button	This button when clicked will open a new page and use native maps to show you a map of La Trobe.		

Settings button	This button when clicked will open a new page and display settings that you can turn on and off.		
Building button	This button when clicked will open a new page and display a list of buildings to choose from.		
Quick Park button	This button when clicked will open a new page and display the best car spot available.		
Car Park Marker	This is used to indicate on the map where the car park are. The whole car park is highlighted.		
Building Marker	This is used to indicate a building on the map. Clicking on the marker will show the building name and the abbreviation.		

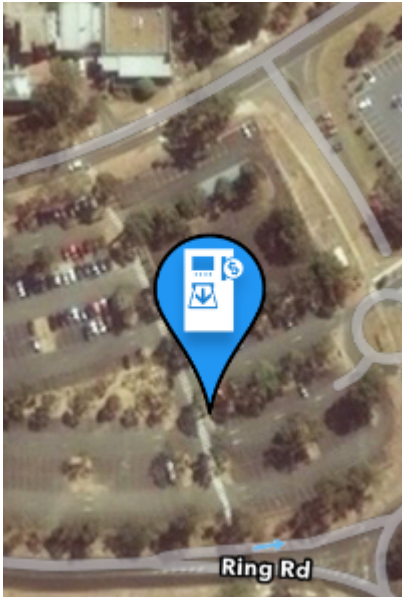
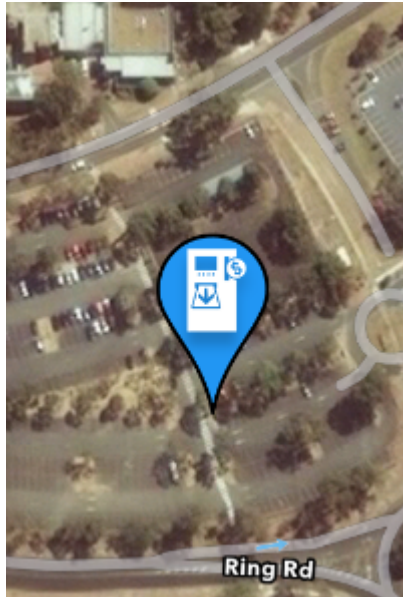
Ticket Marker	This is used to indicate a ticket machine on the map.		
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figure 3.0.1: buttons and icons

4.0 User Manual

[4.1 User Guide \(Android\)](#)

[4.2 User Guide \(iOS\)](#)

4.1 User Guide (Android)

Please read the previous sections of this guide before continuing.

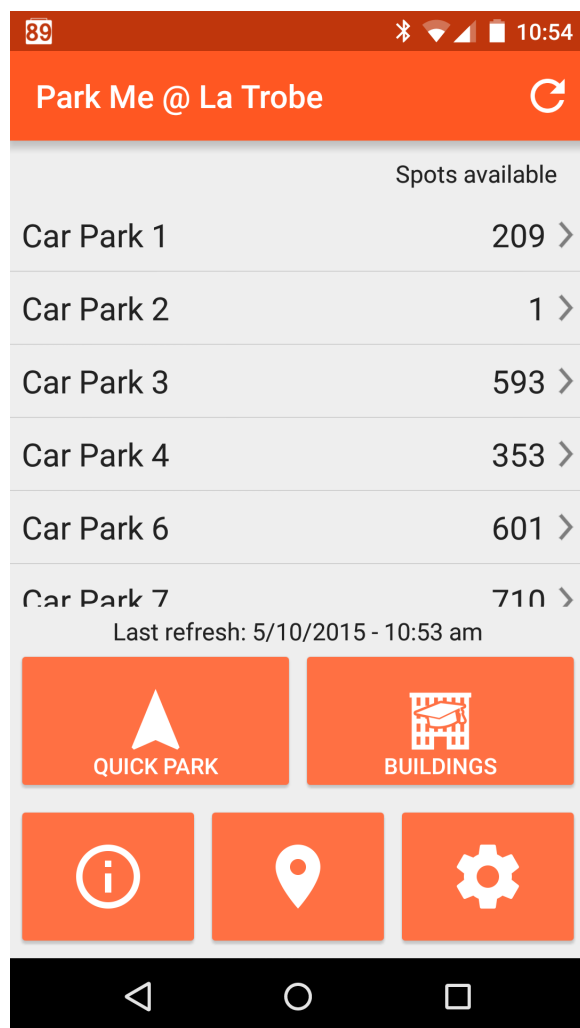
Features Included in this section:

1. Quick links home page
2. Finding a car spot
3. Navigation
4. Information
5. Map

1. Quick Links Home Page:

Use this page to quickly see how many spots are free in each car park.

When the app opens it will open to the quick links home page. It displays a list of the most popular car parks for students and visitors. This list also shows information on the number of available spots in each car park.



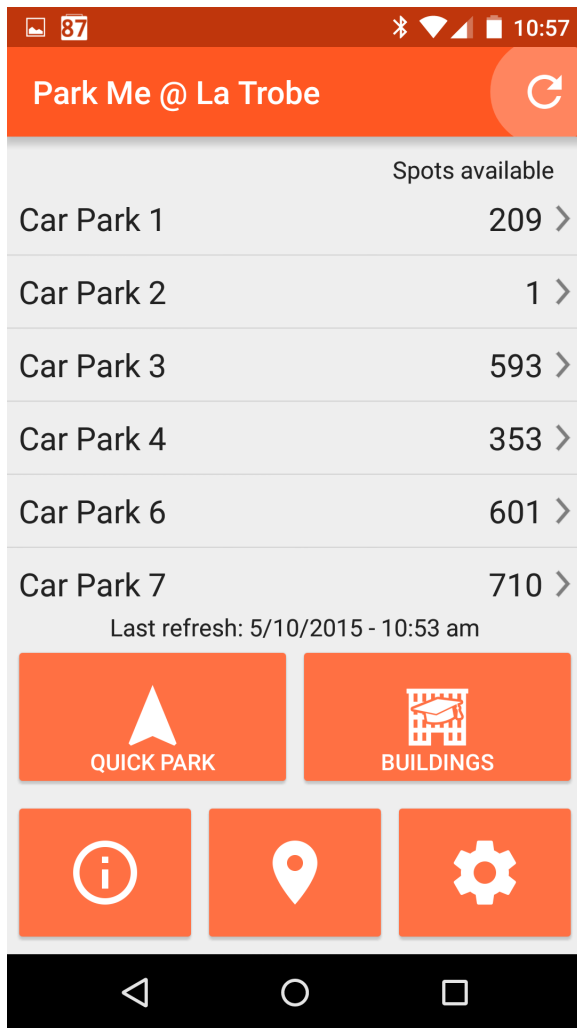


Figure 4.1.1 Home page

Figure 4.1.2 Refresh button on home page



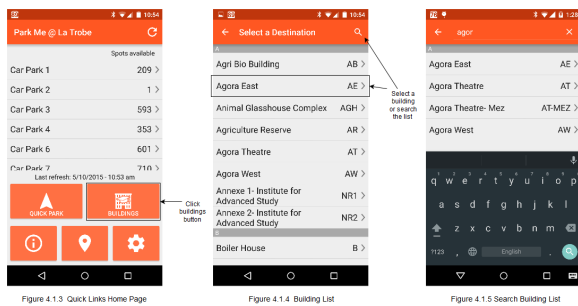
Using the  button to refresh the list will keep the list up - to - date. The time stamp also shows when it was last updated so you can see when you last successfully refreshed.

There are 5 buttons on the page for different things. To navigate to these pages simply tap the buttons. To return to this main page tap the back button in the top left corner. The home page doesn't have a back button because there are no previous pages.

2. Finding a car spot:

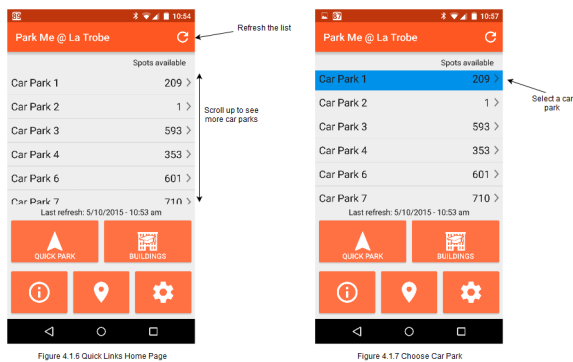
There are three ways to find a car spot: quick park, via building or via car park

1. If you want to park near a building e.g. Beth Gleeson
 - a. Go to the home page
 - b. Tap buildings button
 - c. Scroll through the list until you find the building you need. The list is in alphabetical order. Alternatively use the search feature by clicking the button
 - d. Tap the building you wish to park near
 - e. A new page will open showing you the spot that has been selected. For more information go to feature 3 - Navigation.



2. If you want to park in a certain car park e.g. car park 3

- Go to the home page
- Scroll the car park list to find the car park you want
- Tap the car park you would like to park in,
 - please note if the car park has 0 spots left you will be notified.
- A new page will open showing you the spot that has been selected. For more information go to feature 3 - Navigation.



3. If you to park anywhere on campus and don't care which spot is selected

- Go to home page
- Tap Quick Park
- The best car spot the app can find will be selected for you.
- A new page will open showing you the spot that has been selected. For more information go to feature 3 - Navigation.

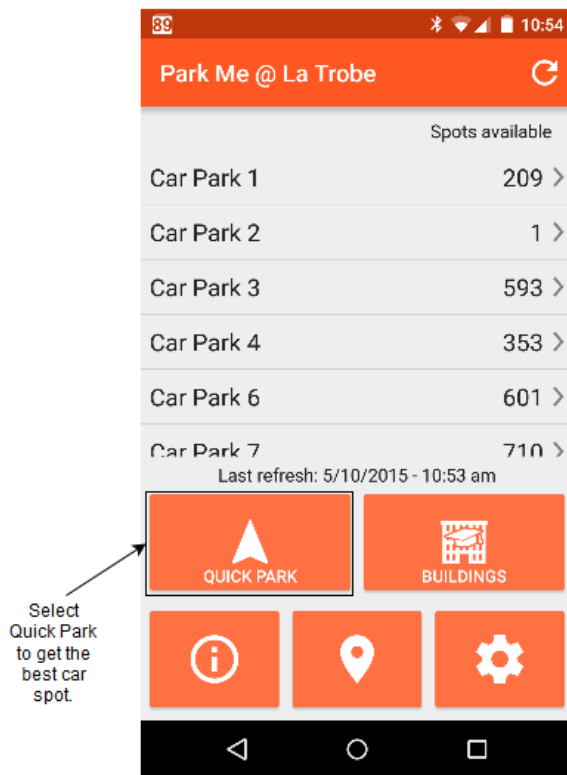


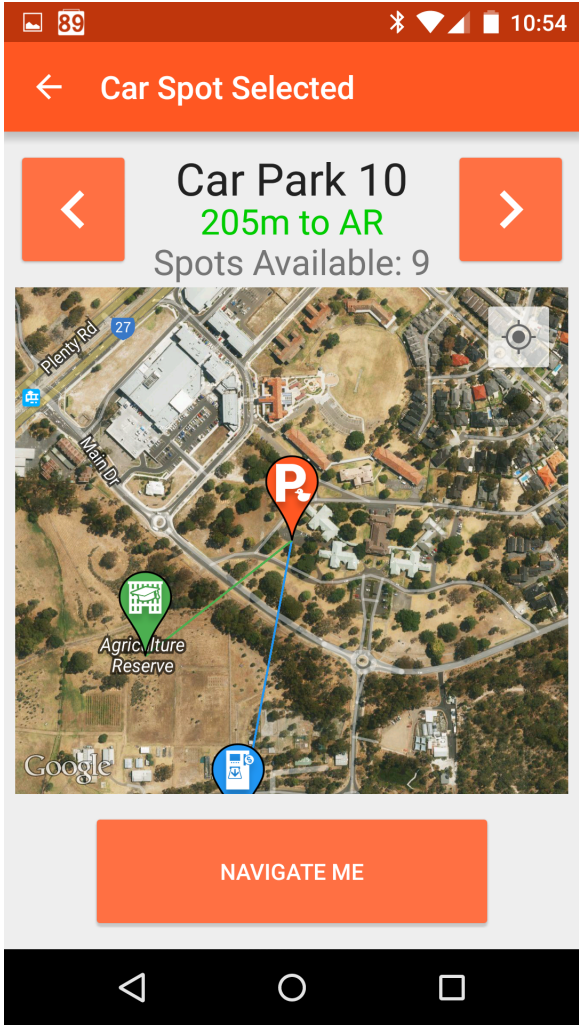
Figure 4.1.8 Quick Links Home Page

3. Navigation

After selecting a Building or Car Park or Quick Park, The app will select the best spot available based on your choice.

When a spot is found you will see a page that has:

- The car park the spot is in.
- The distance to the nearest ticket machine (unless setting is turned off)
- The distance to the building selected (if chose by building)



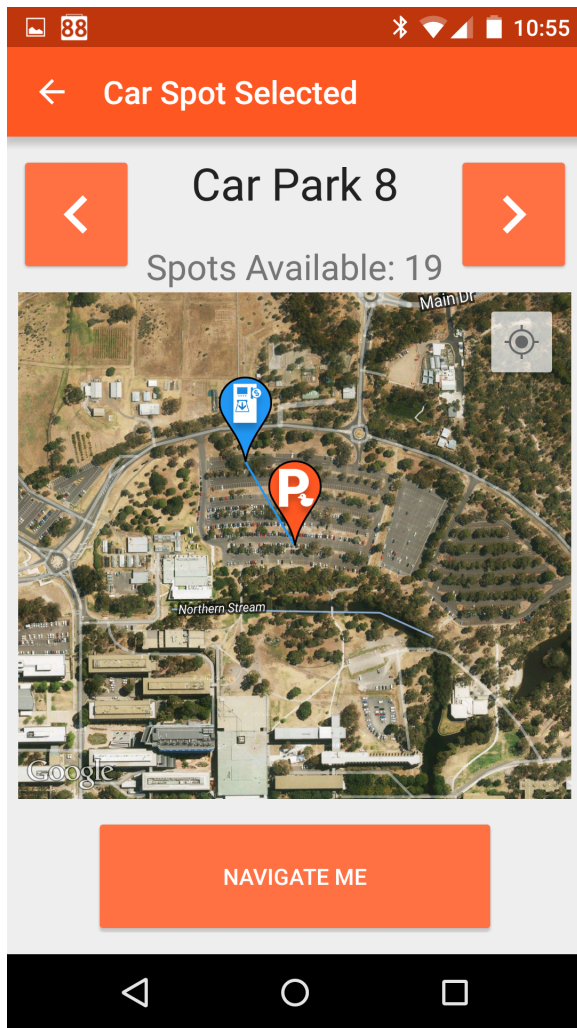


Figure 4.1.9 Selected Car Park (with building)
(without building)

Figure 4.1.10 Selected Car Park

There are markers to indicate where the spot is, where the building is and where the ticket machine is. For more details on the markers please see the getting started guide.

If you wish to launch maps app click the Navigate Me. Launching maps will provide you will step by step voice directions to get to the free car spot while your driving.

4. Information:

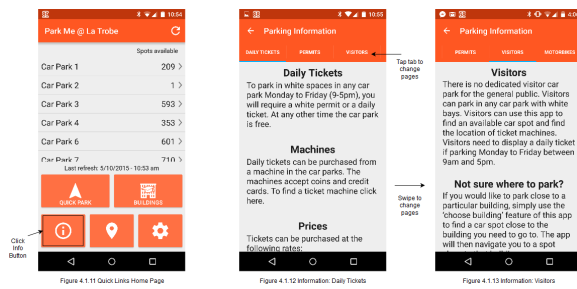
It's is your responsibility as a driver to be aware of the car park rules regarding buying a ticket or permit, where to park and what the rules are.

However to help you understand these rules some information has been provided for you.

To view the information:

1. Open the app
2. Select info
3. Select the tab on information you need. There are 4 tabs. Swipe left 4 times to get to motorbikes.
 - a. Daily Tickets
 - b. Permits
 - c. Visitors
 - d. Motorbikes

4. Use the back button to return to the quick links home page

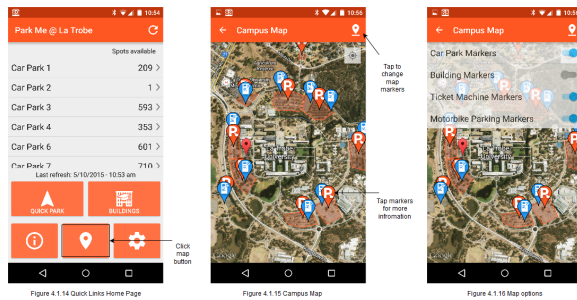


5. Maps

A Campus map as been provided in this app to help you navigate by foot to your class.

Some features of the map include car park numbers, locations of ticket machines, building locations and motorbike parking.

These features can be turned off by tapping the button in the top right corner. Use the sliders to toggle the feature on and off.



4.2 User Guide (iOS)

Please read the previous sections of this guide before continuing.

Features Included in this section:

1. Quick links home page
2. Finding a car spot
3. Navigation
4. Information
5. Map

1. Quick Links Home Page:

Use this page to quickly see how many spots are free in each car park.

When the app opens it will open to the quick links home page. It displays a list of the most popular car parks for students and visitors. This list also shows information on the number of available spots in each car park.

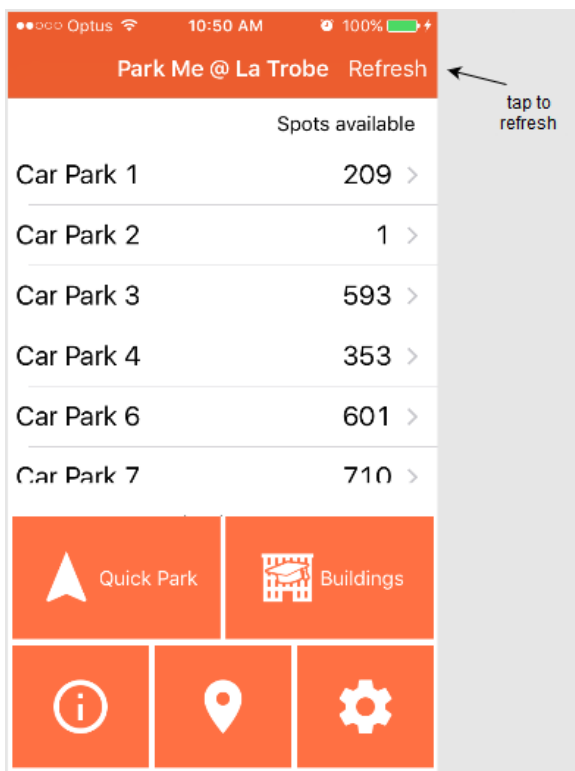


Figure 4.2.1 Home page

Using the refresh button to refresh the list will keep the list up - to - date. The time stamp also shows when it was last updated so you can see when you last successfully refreshed.

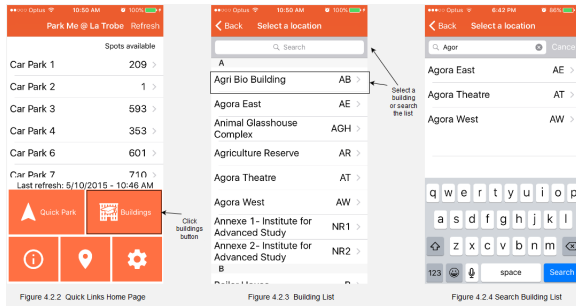
There are 5 buttons on the page for different things. To navigate to these pages simply tap the buttons. To return to this main page tap the back button in the top left corner. The home page doesn't have a back button because there are no previous pages.

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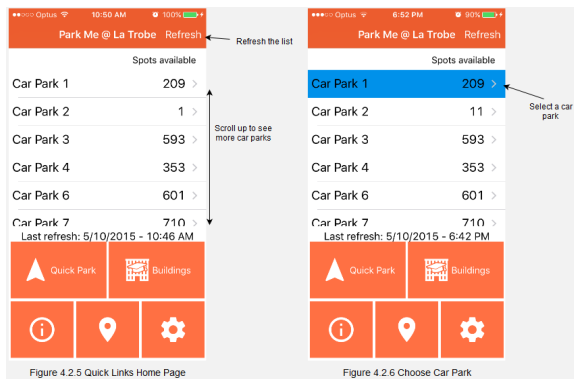
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 - b. Tap buildings button
 - c. Scroll through the list until you find the building you need. The list is in alphabetical order. Alternatively use the search feature to find the building.
 - d. Tap the building you wish to park near

- e. A new page will open showing you the spot that has been selected. For more information go to feature 3 - Navigation.



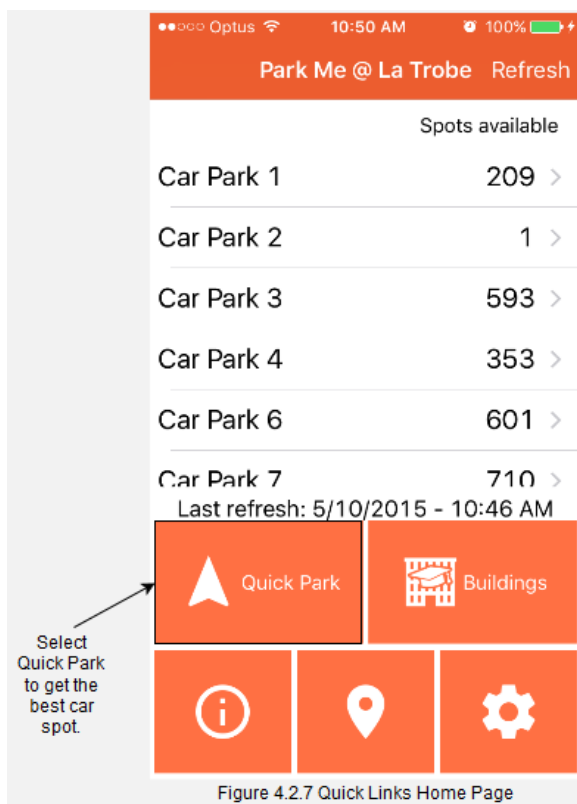
2. If you want to park in a certain car park e.g. car park 3

- a. Go to the home page
- b. Scroll the car park list to find the car park you want
- c. Tap the car park you would like to park in,
 - i. please note if the car park has 0 spots left you will be notified.
- d. A new page will open showing you the spot that has been selected. For more information go to feature 3 - Navigation.



3. If you to park anywhere on campus and don't care which spot is selected

- a. Go to home page
- b. Tap Quick Park
- c. The best car spot the app can find will be selected for you.
- d. A new page will open showing you the spot that has been selected. For more information go to feature 3 - Navigation.



3. Navigation

After selecting a Building or Car Park or Quick Park, The app will select the best spot available based on your choice.

When a spot is found you will see a page that has:

- The car park the spot is in.
- The distance to the nearest ticket machine (unless setting is turned off)
- The distance to the building selected (if chose by building)

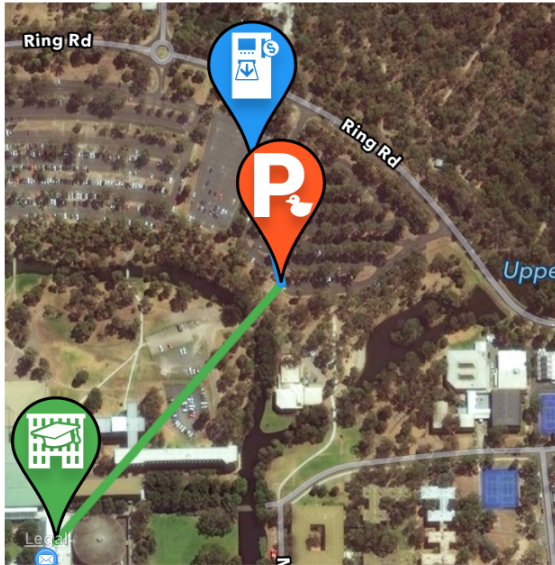
< Select a location Found a car spot!



Car Park 7
300m to AE



Spots Available: 21



Navigate Me

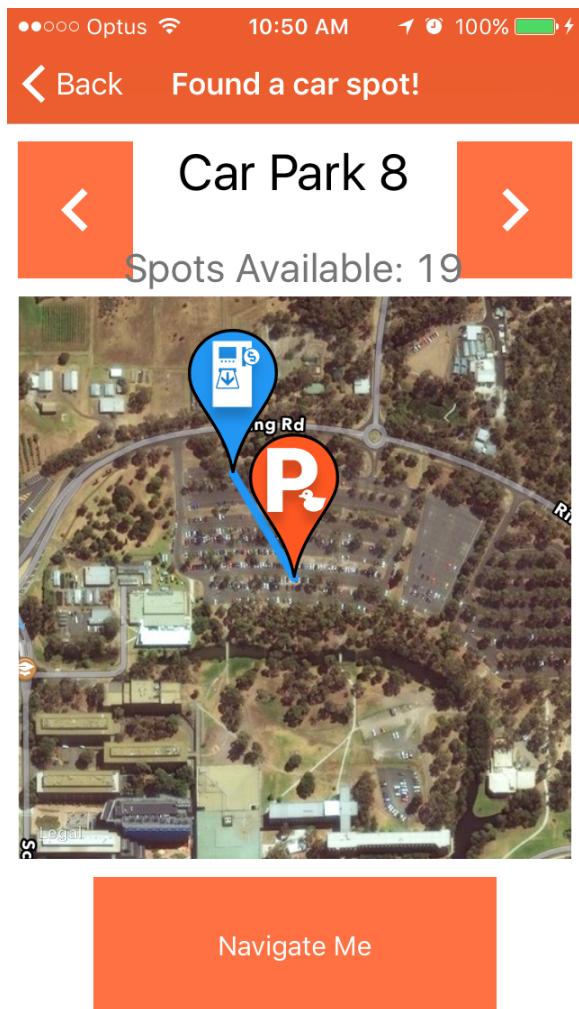


Figure 4.2.8 Selected Car Park (with building)
(without building)

Figure 4.2.9 Selected Car Park

There are markers to indicate where the spot is, where the building is and where the ticket machine is. For more details on the markers please see the getting started guide.

If you wish to launch maps app click the Navigate Me. Launching maps will provide you will step by step voice directions to get to the free car spot while your driving.

4. Information:

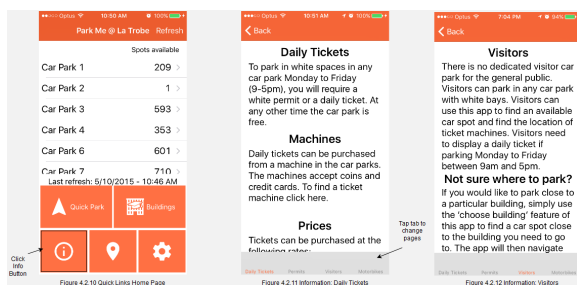
It's your responsibility as a driver to be aware of the car park rules regarding buying a ticket or permit, where to park and what the rules are.

However to help you understand these rules some information has been provided for you.

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3. Select the tab on information you need. There are 4 tabs. Swipe left 4 times to get to motorbikes.
 - a. Daily Tickets
 - b. Permits
 - c. Visitors
 - d. Motorbikes

4. Use the back button to return to the quick links home page

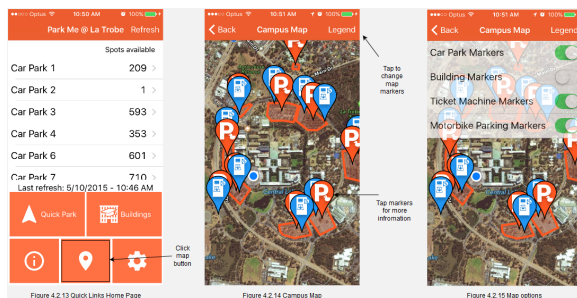


5. Maps

A Campus map as been provided in this app to help you navigate by foot to your class.

Some features of the map include car park numbers, locations of ticket machines, building locations and motorbike parking.

These features can be turned off by tapping the button in the top right corner. Use the sliders to toggle the feature on and off.



5.0 System Administrator Guide

Admin features:

- View log files generated by the app and the hardware nodes
- To view useful statistics on car park behaviour.

View Log Files

An admin can view the log files on the following website: <http://homepage.cs.latrobe.edu.au/15pra02/logFilesWebPage/index.php>. An internet connection is needed.

Admins can use the tabs to view different data. The tab that is currently open is coloured in deep orange. A brief description of the information displayed is given. All data is taken from the database at the time the page was refreshed.

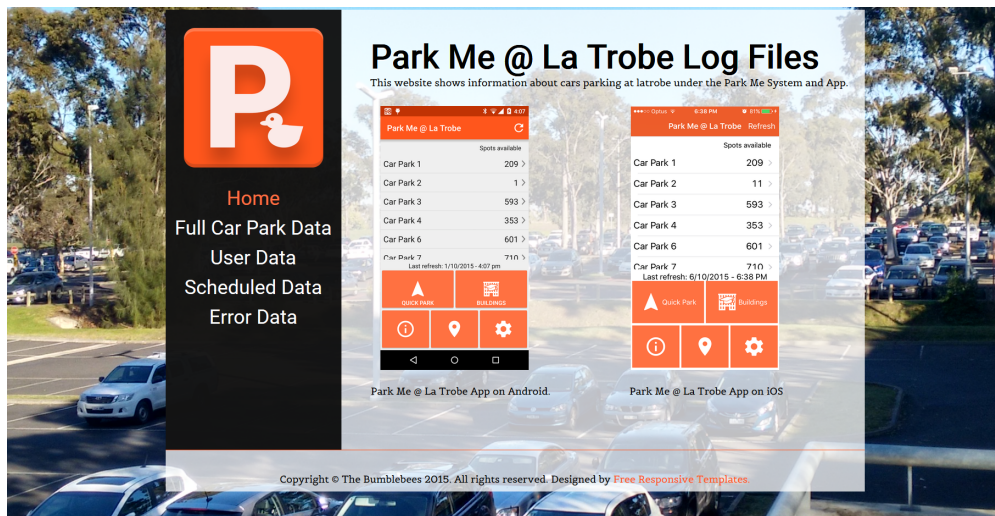


Figure 5.0.1 Database log files web page

Admins can hover the mouse over the graph to view more information about each bar:

Graphs:

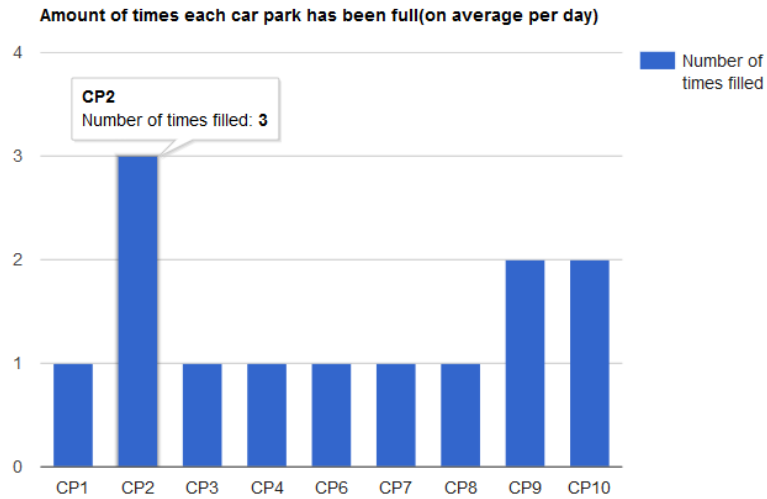


Figure 5.0.2 hover over bar CP2 example.

Admins can use the next and prev buttons to view more raw data. Not all the data can be displayed on one web page and the buttons make it possible to view more data.

Raw Data:

< prev

next >

Filled_id	Car_Park_id	Filled_At	Capacity
1	CP1	2015-09-05 15:46:14	100
2	CP9	2015-09-05 15:47:06	169
3	CP10	2015-09-05 15:47:26	42
4	CP9	2015-09-21 11:16:32	169
5	CP2	2015-09-25 16:36:24	481
6	CP3	2015-09-25 16:36:53	593
7	CP4	2015-09-25 16:37:17	353
8	CP6	2015-09-25 16:37:37	601
9	CP7	2015-09-25 16:37:48	710
10	CP8	2015-09-25 16:37:59	663
11	CP10	2015-09-25 16:38:14	42
12	CP2	2015-09-29 14:09:19	481
13	CP2	2015-09-29 14:24:06	481

Figure 5.0.3 Next and Previous buttons.

Admins can also see useful statistics generated from the data.

Statistics:

The most popular car park of all time is: 3

The most popular car park of this month is: 2

The most popular time of day for car parks to fill up is: 10-11am

Figure 5.0.4 Statistics.

6.0 Installation Guide

This section contains information on how to:

[6.1 Install App](#)

[6.2 Install Hardware Node](#)

6.1 Install App

Android User

Before commencing it is highly recommended to have your device up to date and running the latest version of Android available to you. Make sure you have a Google account and are signed in on your device.

1. Open the Play Store
2. Tap on the search bar at the top of the app.
3. Type: Park Me
4. Select the app and tap INSTALL

Alternative methods of installation are available on Android but will not be discussed in this document. The APK can be made available on request.

To get the latest version of Android, follow the steps in this guide: <http://www.pcadvisor.co.uk/how-to/google-android/how-update-android-smartphone-or-tablet-3614890/>

iOS User

Before commencing it is highly recommended to have your device up to date and running the latest version of iOS available to you. Make sure you have an iTunes account and are signed in on your device.

1. Open the App Store
2. Tap on the search tab at the bottom of the app.
3. Type: Park Me
4. Select the app and tap GET

To get the latest version of iOS, follow the steps provided on the apple website: <https://support.apple.com/en-au/HT204204>

6.2 Install Hardware Node

Contents of this page:

- Install Software on Raspberry Pi
 - Copy working image to raspberry pi
 - Update Operating System before copying (optional)
 - Create image from working node
 - Write saved image to new node
 - Manually Install software
- Setup Detection Software
 - Configure detection settings for individual spots
 - Define a region for spot detection

Install Software on Raspberry Pi

The easiest method to get the software running on a raspberry pi node is to simply make an image of an already working node and then flash this image to another Raspberry Pi ([Win32DiskImager](#) software works well for this). As all the software required to run the Raspberry Pi including it's firmware, kernel and operating system is found on the SD card.

Copy working image to raspberry pi

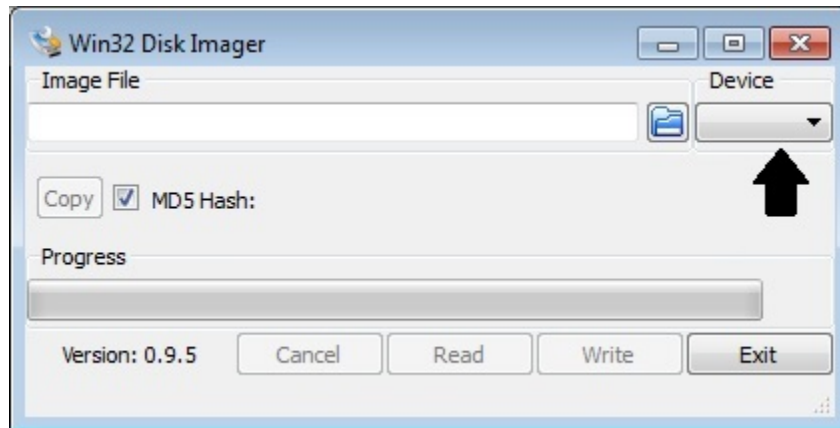
Update Operating System before copying (optional)

It is recommended to try updating all the software and firmware on the Raspberry Pi if writing an image to a newer version of the Raspberry Pi board (for instance copying over from an original Raspberry Pi to a Raspberry Pi 2, or maybe to a Raspberry Pi 3 if upgrading in the future). This can be done by issuing the following commands in a terminal

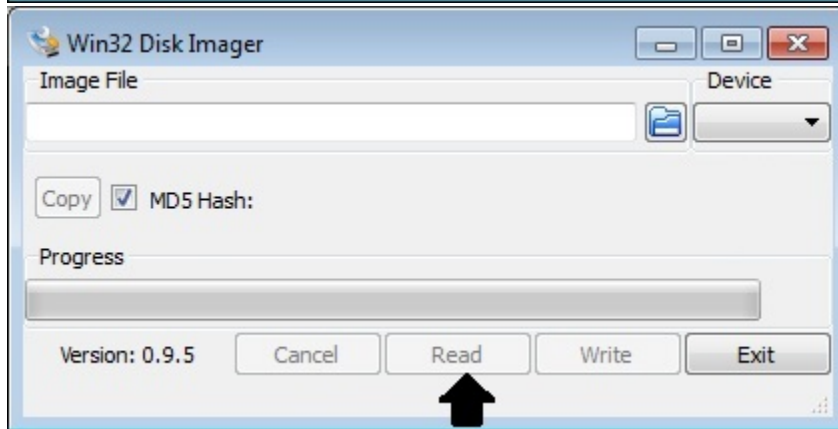
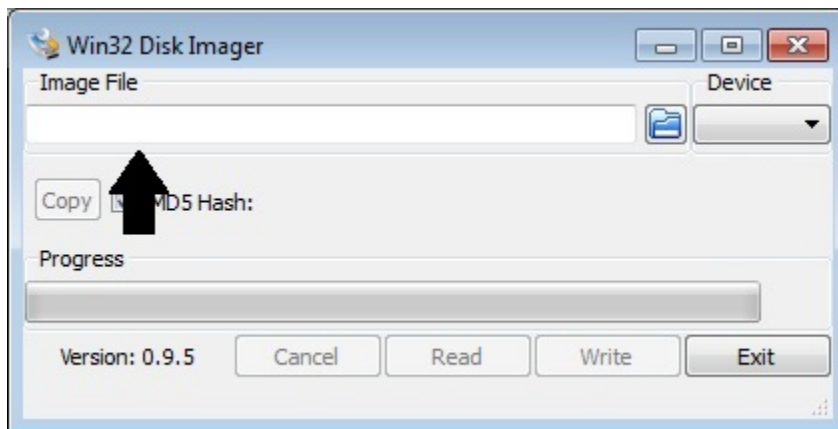
```
sudo apt-get update
sudo apt-get upgrade
sudo apt-get dist-upgrade
sudo apt-get install raspberrypi-ui-mods
```

Create image from working node

Take out the SD card from the working node and insert it into a computer. Run the Win32DiskImager software and select the correct device letter that corresponds to the SD card (check My Computer if unsure).

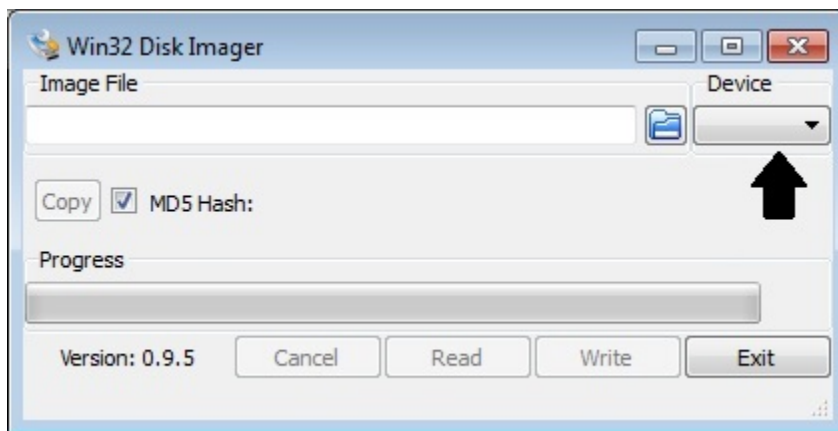


Select where you want to save the image file on your computer and hit the read button.

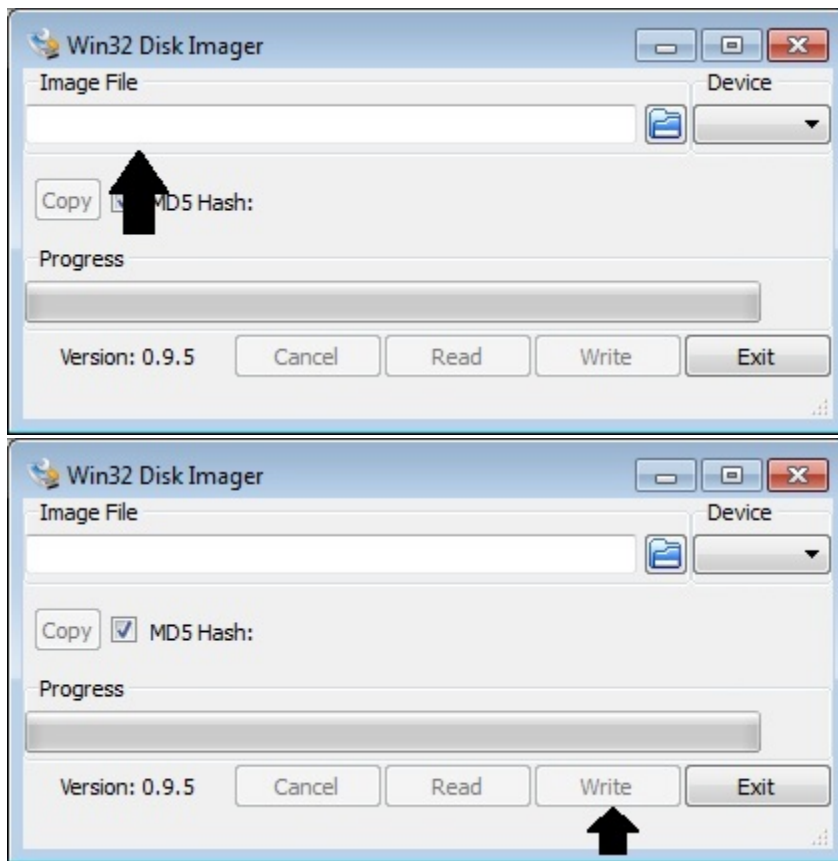


Write saved image to new node

To write a saved image insert the SD card from the new node into a computer. Run the Win32DiskImager software and select the correct device letter that corresponds to the SD card (double check drive letter as this is a destructive operation and will override selected device).



Select which image file on your computer that you want to load and hit the write button.



Manually Install software

Alternatively, If copying and writing the image to the raspberry pi doesn't work then the raspberry pi will have to be flashed with a clean Raspian Image which can be found on the raspberry pi [website](#). Then the following script can be run to install required software packages.

```

#set keyboard to US
sudo sed -i 's/gb/us/g' /etc/default/keyboard
#need to change timezone, hostname, username, password for pi

#Install dependencies
sudo apt-get -y install build-essential git cmake pkg-config libjpeg8-dev libtiff4-dev libjasper-dev
libpng12-dev libavcodec-dev libavformat-dev libswscale-dev libv4l-dev libgtk2.0-dev libatlas-base-dev
gfortran python2.7-dev python3-dev && cd ~ && git clone https://github.com/Itseez/opencv.git && cd
opencv && git checkout 3.0.0 && cd ~ && git clone https://github.com/Itseez/opencv_contrib.git && cd
opencv_contrib && git checkout 3.0.0 && wget https://bootstrap.pypa.io/get-pip.py && sudo python3 get-
pip.py && sudo pip3 install --no-cache-dir matplotlib virtualenv virtualenvwrapper PyMySQL && sudo rm -
rf ~/.cache/pip && sudo pip install pytest
#Setup and compile opencv
echo "# virtualenv and virtualenvwrapper" >> ~/.profile && echo "export VIRTUALENVWRAPPER_PYTHON=/usr/bin
/python3" >> ~/.profile && echo "export WORKON_HOME=$HOME/.virtualenvs" >> ~/.profile && echo "source
/usr/local/bin/virtualenvwrapper.sh" >> ~/.profile && source ~/.profile && mkvirtualenv cv3 && workon
cv3 && pip install numpy && cd ~/opencv && mkdir build && cd build && cmake -D CMAKE_BUILD_TYPE=RELEASE
-D CMAKE_INSTALL_PREFIX=/usr/local -D INSTALL_C_EXAMPLES=ON -D INSTALL_PYTHON_EXAMPLES=ON -D
OPENCV_EXTRA_MODULES_PATH=~/opencv_contrib/modules -D BUILD_EXAMPLES=ON -D WITH_QT=ON .. && sudo
make -j4 && sudo make install && sudo ldconfig && cd ~/.virtualenvs/cv3/lib/python3.2/site-packages/ &&
ln -s /usr/local/lib/python3.2/site-packages/cv2.cpython-32mu.so cv2.so

#wifi Gui
sudo apt-get install wpagui raspberrypi* raspi-config && sudo rm /etc/network/interfaces && echo "auto lo
iface lo inet loopback

auto eth0
allow-hotplug eth0
iface eth0 inet dhcp

auto wlan0
allow-hotplug wlan0
iface wlan0 inet manual
wpa-roam /etc/wpa_supplicant/wpa_supplicant.conf
iface default inet dhcp
" | sudo tee -a /etc/network/interfaces && echo "ctrl_interface=DIR=/var/run/wpa_supplicant GROUP=netdev
update_config=1" | sudo tee -a /etc/wpa_supplicant/wpa_supplicant.conf && sudo chmod 600 /etc
/wpa_supplicant/wpa_supplicant.conf && sudo adduser pi netdev && sudo ifdown wlan0 && sudo ifup wlan0

#mesh network software
git clone https://github.com/urlgrey/hsmm-pi.git && cd hsmm-pi && sh install.sh

```

Setup Detection Software

Configure detection settings for individual spots

Define a region for spot detection

The detection region defines where a spot lies. If the detection algorithm detects something within this region the spot is marked as taken. The region is a Polygon with each point in the polygon defined from the x,y coordinates from the cameras field of view. (with the top left corner of the camera being (0,0) and the bottom right being the resolution of the camera e.g. (1024,720))

7.0 Special Notes

This section contains the following pages:

[APPENDIX_a Statement of authorship](#)

[Bibliography](#)

[Glossary of Terms](#)

APPENDIX_a Statement of authorship

Statement of Authorship

Department of Computer Science and Computer Engineering

STATEMENT OF AUTHORSHIP FOR GROUP

ASSIGNMENTS

Please read the following instructions and associated University documents before completing and submitting this form with your item of assignment.

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2. The Academic Misconduct Policy found at <http://www.latrobe.edu.au/policies>
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4. Regulation 16.2 found at [<http://www.latrobe.edu.au/legalservices/regulations.html> | <http://www.latrobe.edu.au/legalservices/regulations.html>]

We each certify that our own contribution in the attached material is original work and that we have no knowledge of an act of plagiarism committed by any member of this group. We declare that no other person's work has been used without due acknowledgement. Except where we have clearly stated that we have used some of this material elsewhere, it has not been presented by us for examination in any other course or unit at this or any other institution.

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We understand the nature of plagiarism to include the reproduction of someone else's words, ideas or findings and presenting them as our own without proper acknowledgement. Further, we understand that there are many forms of plagiarism which include direct copying or paraphrasing from someone else's published work (either electronic or hard copy) without acknowledging the source; using facts, information and ideas derived from a source without acknowledgement; producing assignments (required to be independent) in collaboration with and/or using the work of other people; and assisting another person to commit an act of plagiarism. We understand that the work submitted may be reproduced and/or communicated by the University or a third party authorised by the University for the purpose of detecting plagiarism.

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Lecturer/Tutor's Name**Dr Torab Torabi**..... Assignment No.**1**.....

Name of Document**User Document**.....

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Given Names.....Kylie Maree	Student No17683457
SignatureKylie Maree Marshall	Date29/05/15
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2. Paraphrasing someone else's words without acknowledging the source;
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5. Producing assignments which should be the student's own, independent work in collaboration with and/or using the work of other people (e.g. a student or tutor).

Assisting another person to plagiarise material may be punished as severely as is plagiarism itself. Assisting plagiarism may involve a student lending work (or by posting it on the Internet for sale) which is intended for submission for assessment, or which has already been submitted, so that it can be copied and handed in by another student as that student's own work.³

Students may find it helpful to discuss assignments with other students and their tutors. The University encourages students to communicate with one another in constructive ways about the learning process. Students may choose to assist each other, for example in discussing the approaches that might be taken to assignment topics, or helping with the availability of reading materials. They should, however, write their assignments independently, except when they are asked to work on a project as a member of a group which is to submit a joint report, as equal contributors. In this case, except where the final written work consists of sections for which particular individuals take sole responsibility, the group as a whole assumes responsibility for it and the proper acknowledgment of any use made of the words or ideas of people outside the group.

The increased vigilance of the University in regard to plagiarism is in response to the perceived growing incidence of plagiarism. The new recommended penalties reflect the value the University places on academic honesty and how seriously it regards plagiarism offences. The University will protect its reputation for academic integrity by ensuring that serious and serial offenders, who wish to obtain an unfair advantage through cheating, are given penalties proportionate to the offences committed.

A summary of the current legislation is attached in Appendix A. This legislation will be amended if the Working Party's recommendations are approved.

Bibliography

Glossary of Terms

A

Android - Android is a popular Phone and Tablet operating system developed by Google.

API - Application Programming Interface is a collection protocols, routines and other instructions that grant inter-application level access for building applications.

App - Shorten popular colloquial language for "Application". A piece of software that exists at the operating system level.

B

Backend - Supporting program functionality that works behind the scenes without user interaction.

C

Camera-node - Synonymous with "Hardware-Node" and "RaspberryPi Node", A camera-node is the collective hardware that collects carpark information for the collective car park system.

Cross-platform - a compatibility with different systems separated by operating systems.

D

[Draw.io](#) - An online application used for capable of making various types of diagrams using such standards as UML.

F

Flag - A binary value used to indicate a state for something.

Float - A datatype for holding numerical information.

G

GPS - Global Positioning System is a coordinate device used to generate positioning data for use on a map.

GPS-Navigation - A system for navigating on a map.

GUI - Graphical User Interface is a graphical representation of various interactive elements for the user to navigate functionality.

H

Hardware-Node - Synonymous with "Camera-Node" and "Raspberry Pi Node", A Hardware-node is the collective hardware that collects carpark information for the collective car park system.

HTTP - Hypertext Transfer Protocol is the protocol responsible for transmitting information across the web.

I

iOS - iOS is a popular Phone and Tablet operating system developed by Apple.

J

JavaScript - JavaScript is a programming language primarily used in creating interactive web elements.

Json Object - JavaScript Object Notation Object is apart of the JavaScript language as an easy to read formatted object.

M

Mesh Network - Mesh Network is a network Topology in which Network nodes act as relays for network data in the network.

R

Raspberry Pi Node - Synonymous with "Hardware-Node" and "Camera Node", A Raspberry Pi node is the collective hardware that collects carpark information for the collective car park system.

Runtime - The period of time in which a software is running.

S

SDK - Software Development Kit is a tools used for the aid in developing software for a certain platform.

String - A software datatype which consists of a finite sequence of characters.

U

UML - Unified Modelling Language is a modelling language used to universally model system design structure for programming languages.

User Story - An agile tool for describing in the view of a user a requirement for a project.

W

Wi-Fi - A wireless computer networking technology used to allow electronic devices to communicate with each other.

Wireframe - A schematic for different visual elements in a Graphical User Interface.